

# Chef Recipes to Configure the Desired State of a System



Chef is the answer when managing dynamic infrastructure. It automates how applications are configured, deployed and managed across your network, irrespective of its size. This is the fourth article in our series on Chef. Read on for better insights into this versatile tool.

In the previous articles in the DevOps series, we discussed the procedure to set up a client-server sandbox environment as well as the Chef client and server installation. In this article, we are going to talk about Chef's architecture and the basics and internals of a cookbook. A clear understanding of the Chef architecture and its internals will help view Chef in a holistic way.

## Chef's architecture

Figure 1 gives an overview of the Chef architecture. An organisation is independent of the components of Chef. It can be applied to your business unit or specific domain or project account. The DevOps process involves maintaining a database, source control management, software development, as well as testing the developed software and deployment in the production environment. Continuous integration (CI) plays a vital role here to invoke the execution of new checked-in code to the next stage. For example, if any code is checked-in by the developer, the CI triggers the build process, then tests the code and, finally, enables the deployment. If the code fails at any stage, it will alert the concerned stage and respective team requesting it to look into the matter. So an organisation

can have different environments for development, testing and production. Further, a development environment can have different roles like maintaining software code and the relevant data in a database server, executing the application in the application server and monitoring the application by a dedicated monitoring server. Each role may associate one or more client nodes and every node can have one or more cookbooks. Nodes denote servers in the infrastructure, while the servers may be physical systems or virtual machines or hardware, which will represent compute instances. A cookbook consists of recipes, which in turn give the *run* list to enable the given nodes to reach a desired state after the deployment process. It may be creating a Web server, an *ftp* server, a mail server, a specific registry key, creating files or directories, manipulating file systems, setting up network components, installation of software packages, automating frequently executed tasks and managing services, etc.

So, to use Chef to configure your infrastructure, you should have a fair knowledge of cookbooks. This article will describe the writing of recipes in a cookbook. A recipe is about managing different resources and a resource is a fundamental building block of Chef. Let us name the cookbook